

SUMMARY

I am a Master's student in Computer Science at [Dartmouth College](#) building machine learning systems for ecological monitoring and climate risk decision-making. My work combines computer vision, sensing systems, and uncertainty-aware modeling to generate actionable insights for biodiversity tracking and climate adaptation. I previously worked in the industry for 6+ years as a Machine Learning specialist building computer vision, NLP, and decision-support pipelines.

Research focus: ML for biodiversity monitoring, ecological sensing, and decision-making under climate uncertainty.

EDUCATION

New Hampshire, USA	Dartmouth College	Sept. 2025 - Present
▪ M.S. in Computer Science ▪ Coursework: Artificial Intelligence, Visual Computation, Machine Learning, Natural Language Processing (NLP), Computational Photography, Robotics, Environmental Data Science, Climate Science and Engineering, Oceanography. ▪ Teaching Assistant: NeuroSymbolic AI, Machine Learning and Statistics, Computational Photography		
New Delhi, India	IIIT-Delhi	Sept. 2016–Jun. 2020
▪ B.Tech. in Computer Science, Minor in Economics ▪ Coursework: Programming Languages, Data Structures & Algorithms, Operating Systems, Natural Language Processing (NLP), Machine Learning (ML), Image Processing, Multimedia Computational Analysis, Game Theory, Econometrics, Combinatorics, Finance, Value Portfolio		

PROJECTS

Blacklight-Sheet Imagery for Tracking Trends in Lepidoptera (Moth) Abundance **Sept 2025–Present**

Advisors: [Dr. Matt Ayres](#), [Dr. David Lutz](#) with Hubbard Brook Experimental Forest ([HBEF](#))

- Enabled non-lethal, scalable biodiversity monitoring as an alternative to traditional trapping-based methods.
- Published a reproducible ecological imagery [dataset](#) on Environmental Data Initiative [EDI](#). [\[Code\]](#)
- Developed a computer vision pipeline to estimate ecological abundance from field imagery, converting detections into standardized biodiversity indices for temporal ecological analysis.
- Transformed biodiversity indices into biomass distributions using size-bin classification.
- Processed 20k+ images across five ecological sites to generate multi-year biodiversity time series.

Utilized: Python, OpenCV, Object Detection, Non-Maximum Suppression, Time-Series Analysis, Ecological Field Data

Adaptive Flood Planning Under Deep Uncertainty

Spring 2026–Present

Advisor: [Dr. Klaus Keller](#) | Dartmouth College

- Developing decision-support workflows for climate adaptation under flood risk, with a focus on uncertainty-aware planning and adaptive policy design.
- Implementing uncertainty-aware decision frameworks like multi-objective optimization, policy evaluation across scenarios.
- Framing long-term flood adaptation as a sequential decision problem by incorporating uncertainty, timing, and cost into policy pathways.
- Building upon existing flood-risk and decision-analysis literature to evaluate robust and adaptive planning strategies.

Utilized: Python, Decision Analysis, Uncertainty Quantification, Climate Risk Modeling, Adaptive Planning

Low-Cost Autonomous Bat Monitoring Using Event-Triggered Sensing

Winter 2026

Advisor: [Dr. Alberto Quattrini Li](#) | Dartmouth College

- Designed a system for passive wildlife monitoring, enabling scalable behavioral data collection in low-resource environments.
- Built a wildlife monitoring prototype using Raspberry Pi, PIR sensors, camera modules, and OpenCV-based motion detection for event-triggered video capture.
- Designed a sensing pipeline to record and annotate detections, generating trajectory data for downstream analysis.
- Evaluated tradeoffs in autonomous ecological monitoring under real-world deployment constraints.

Utilized: Python, OpenCV, Raspberry Pi, GPIO, Picamera2, Motion Detection, Ecological Sensing

PUBLICATIONS

- **Suhavi, S.**, Lutz, D.A., Jones, J.S., Ayres, M.P. **2026**. Blacklight Sheet Invertebrate Photos at Hubbard Brook Experimental Forest; 2023. Environmental Data Initiative. [\[DOI\]](#)
Reproducible ecological imagery dataset for non-lethal invertebrate monitoring across Hubbard Brook blacklight-sheet sampling sites.
- **Suhavi**, Singh, A., Arora, U., Shrivastava, S., Singh, A., Shah, R.R., Kumaraguru, P. **2022**. Twitter-STMHD: An Extensive User-Level Database of Multiple Mental Health Disorders. ICWSM. [\[PDF\]](#)
Led a team of 4 researchers to build the largest temporally-contextual MHD dataset, in line with DSM-5 guidelines which built correlations between clinically recognized symptoms and behavioral online attributes.

EXPERIENCE

Global Science AI Intern	World Wildlife Fund	June 2026
▪ Building an automated report-analysis pipeline to classify conservation projects and extract structured insights from historical project documentation. ▪ Developing evaluation metrics for project resilience and reproducibility to support comparative analysis of conservation interventions. ▪ Applying NLP and machine-learning methods to organize project evidence and improve access to institutional conservation knowledge.		

Senior Data Scientist	Scry AI	June 2020-July 2025
Collatio : An Intelligent Document Reader		
▪ Built and led the product's Indian Data Science Team, managing 7 engineers ▪ Designed, developed and managed the document classification pipeline; it uses text & image-based models, SVMs, TF-IDF transformations and LLM-based reconciliation, achieving approximately 99% recall ▪ Developed an ensemble pipeline with LayoutLM, LLM (MistralAI, phi3), and spatial NLP models for entity/value extraction, achieving an accuracy of more than 97% ▪ Devised a precision-first formula for stacking information retrieval models, a foolproof method to achieve over 85% precision, the theory became the foundation for all future ensemble constructions ▪ Introduced and improved upon object detection models for face and logo detection (ArcFace, FaceNet, YOLOv5)		

SKILLS

- **Geospatial & Environmental Data**: Remote sensing, spatial data analysis and ecological time-series
- **Machine Learning**: Supervised and Unsupervised Learning Algorithms (Regression, Random Forest, SVM)
- **Deep Learning**: Object-Detection with YOLO & Faster R-CNN; CNN, LSTM, LayoutLM, BERT, LLMs
- **Techniques**: Data Preprocessing, Feature Engineering, Large-Scale Data Handling (images, texts, videos), ETL Pipelines, hyper-parameter tuning, NLP, Hypothesis Testing, Time-Series Analysis, Classification, Ensemble-Building
- **Programming Languages**: Python, Pyspark, Linux, R
- **Technologies**: TensorFlow, PyTorch, Scikit-Learn, Pandas, Numpy, OpenCV, Git, Flask, Docker, Azure, AWS.

NOTABLE ACHIEVEMENTS

Neukom Fellowship for researching Computational Decisions Making under Deep Uncertainty	June 2026
Winner of the Open & Reproducible Research Award OSCARRs by Dartmouth	March 2026
Best Project Award in COSC 273, Computational Photography	June 2026
Invited as a technical mentor at the First Meta Llama Hackathon in India	November 2024
Placed on the Dean's List for Leadership (IIIT-Delhi)	2018
President of the Entrepreneurship Cell (ECell) IIIT-D Chapter	2019-2020